SE-computer(Div-B)	Roll number : 9947
Experiment no. : 1	
Academic Year :Jan- April 2024	
Related Course outcome : At the end of the course, Students will be able to design ER model and develop relational model	

Rubrics for assessment of Experiment:

Indicator	Poor	Average	Good
Timeliness ● Maintains assignment deadline (3)	Assignment not done (0)	One or More than One week late (1-2)	Maintains deadline (3)
Completeness and neatness ● Complete all parts of ER diagram(3)	N/A	< 80% complete (1-2)	100% complete (3)
Originality ● Extent of plagiarism(2)	Copied it from someone else(0)	At least few questions have been done without copying(1)	Assignment has been solved completely without copying (2)
Knowledge ● In depth knowledge of the assignment(2)	Unable to answer 2 questions(0)	Unable to answer 1 question (1)	Able to answer 2 questions (2)

Assessment Marks :

Timeliness	
Completeness and neatness	
Originality	
Knowledge	
Total	

Total : (Out of 10)

Teacher's Sign :

Name Student		Roll No.	
Lab Experiment No.	1	Date	
Expt. Title	Problem Definition and draw ER /EER diagram		
Tool Used :	draw.io. or Lucidchart		

Aim : Problem Definition and draw ER /EER diagram

Objective of the Experiment:

1. Defining problem statement.

Theory : (Sample Problem statement-Advertising agency)

An "Advertising agency" is a service business dedicated to creating, planning, and handling advertising for its "clients or customers" Here clients are "companies" and that companies produce "products". This product needs print or television or Internet ads in order to increase their sales in the market. An Advertising agency independent from the client and provides an outside point of view to the effort of selling the client's products or services. An agency can also handle overall marketing and branding strategies and sales promotions for its clients. Inside Advertising agency mainly three departments are there. They are

1. CUSTOMER RELATIONSHIP DEPARTMENT

This department takes the orders from clients in order to give ads on their products.

2. FINANCE DEPARTMENT

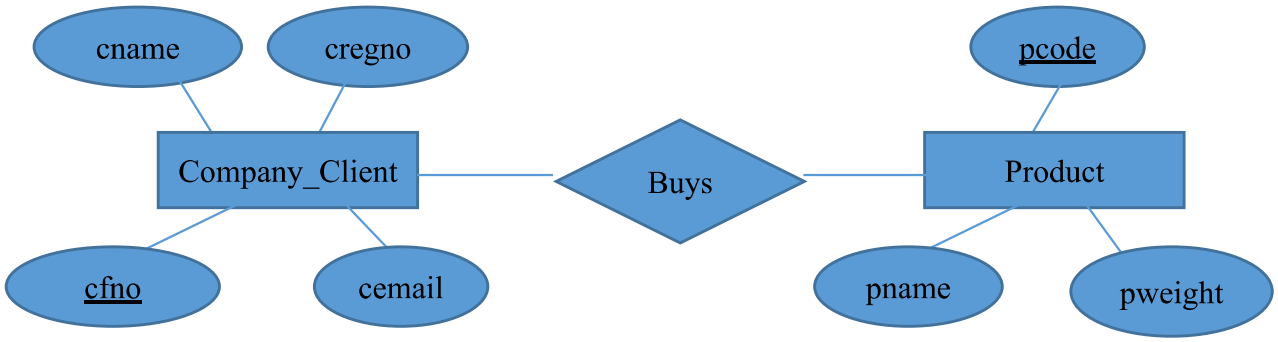
" Finance department " deals with the finance matters between clients and Advertising agency just like how much money client paid at initial time and what is the balance amount and what is the last date to pay balance etc..

3. CREATIVE DEPARTMENT

Creative department is a department where "employees" creates the actual ads from the core of an advertising agency. Creative department frequently work with "productionstudios" to develop and implement their ideas.

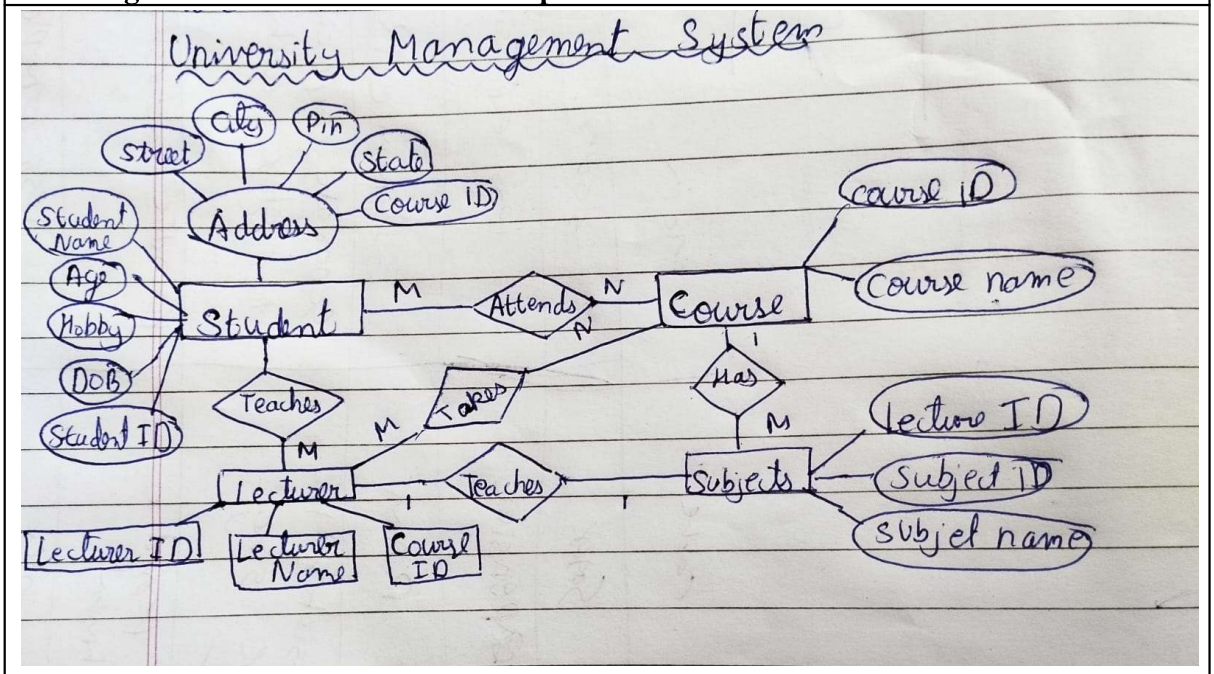
In Ad agency large number of "employees" exists. For example (i) Creative, drawing artists & copy writers work in creative department. (ii) Account manager handles finance department. (iii) .Branch manager handle ad branch etc. for these employees "dependents" may also exists.

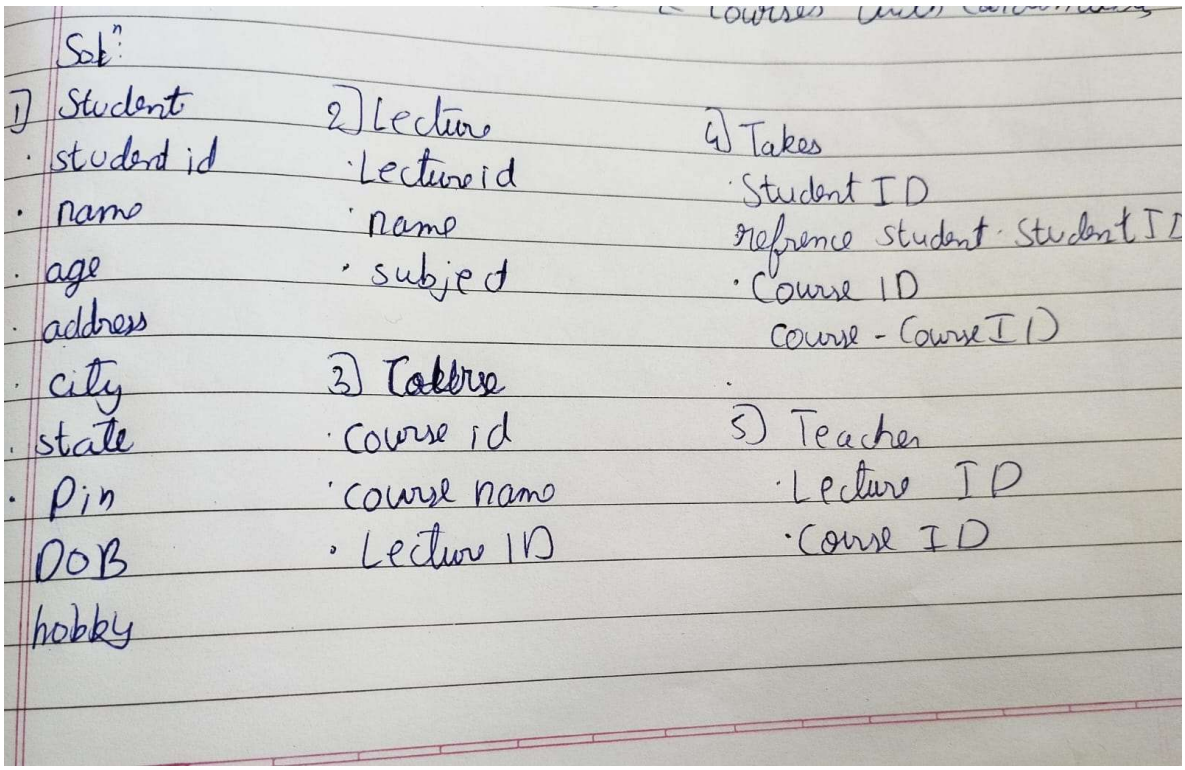
Sample E-R diagram:-



Description of Problem Statement:-

E-R diagram for the Problem to be implemented.





Set:

1) Student
- student id
- name
- age
- address
- city
- state
- Pin
- DOB
- hobby

2) Lecture
- Lecture id
- name
- subject

3) Courses
- course id
- course name
- Lecture ID

4) Takes
- Student ID
  refrence student · Student ID
- Course ID
  course - Course ID

5) Teacher
- Lecture ID
- Course ID

Post Lab Assignment:
1) Describe various symbols used in E-R Diagram

→ Entities.
- Rectangle : Represent real word thing or Concept
→ Attributes.:
- Oval : Shown characteristics of an entity
- Underlined : Indicates the primary key uniquely identifying each entity
→ Relationships.
- Diamond : Represent the connection between entities
- Lines : Connect the diamond to the participating entities
→ Cardinality
- 1:1 → One istance of each enitity relates to one instance of other
- 1:N → One instance of each entity relates to many instance
- N:M → Many istance of each entity relates to many instance of other
- Double-lined rectangle : Represents a weak entity that depends on another entity for instance.
- Dashed lines : Optional participation in a relationship
- Text : Annotation can be added to clarity